10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 9.1 — Tangents, Contact of Circles and Angles

Unit 9 | Solved Exercise | New Book (PECTAA)

Exercise 9.1 (Solved)

Three rules carry this exercise. A tangent is perpendicular to the radius at the point of contact. Two tangents drawn from an outside point are equal in length. The angle at the centre is twice the angle at the circumference.

1. Find the value of x.

In the figure A is an outside point and two tangents from A touch the circle at B and at the second point, with ∠A = 38°. The angle x is ∠AOB.

The radius OB is perpendicular to the tangent AB

∠OBA = 90°

In triangle OBA: x + 90° + 38° = 180°

x = 180° − 128°

x = 52°

Answer: x = 52°

2. Find the angle ABC.

A is an outside point; AB and AC are the two tangents and ∠BAC = 65°.

Two tangents drawn from an outside point are equal

AB = AC, so triangle ABC is isosceles

$$\angle ABC = \angle ACB = \frac{180° - 65°}{2}$$

$$\angle ABC = \frac{115°}{2} = 57.5°$$

Answer: ∠ABC = 57.5°

3. A, B and C are points on the circumference of a circle with centre at O. AC is the diameter of the circle and DE is the tangent to the circle at the point C and m∠BCD = 62°. Find (i) m∠BCA (ii) m∠BAC.

AC is a diameter, so the angle in the semi-circle is 90°

∠ABC = 90°

The angle between the tangent CD and the chord CB equals the angle in the alternate segment

∠BAC = ∠BCD = 62°

In triangle ABC: ∠BCA = 180° − 90° − 62°

∠BCA = 28°

Answer: (i) ∠BCA = 28° (ii) ∠BAC = 62°

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4. D, E and F are points on the circumference of a circle with centre at O and m∠DOF = 130°. Find m∠DEF.

The angle at the centre is twice the angle at the circumference on the same arc

$\angle DEF = \frac{1}{2} \times \angle DOF$

$\angle DEF = \frac{130°}{2} = 65°$

Answer: ∠DEF = 65°

5. X, Y and Z are points on the circumference of a circle with centre at O and m∠XYZ = 85°. Find x.

In the figure x is the reflex angle ∠XOZ at the centre.

Angle at the centre on the minor arc = 2 × ∠XYZ

= 2 × 85° = 170°

The reflex angle completes the full turn

x = 360° − 170°

x = 190°

Answer: x = 190°

6. In a historical monument, a circular fountain with a radius of 3 m is built. A flagpole is erected 7 m away from the centre of the fountain. Two ropes from the pole are tied to the edge of the fountain, just touching it. Find the length of each rope.

A rope that just touches the circle is a tangent, so the radius drawn to the point of contact is perpendicular to the rope.

OP = 7 m, radius r = 3 m

In the right triangle OPT: $PT^2 = OP^2 - r^2$

$PT^2 = 7^2 - 3^2 = 49 - 9 = 40$

PT = √40 = 6.32 m

Answer: Length of each rope ≈ 6.32 m

7. Two circular gears touch each other externally for proper rotation in a machine. The radii of the two circular gears are 5 cm and 7 cm. What is the distance between their centres if they touch externally?

For external contact the distance between the centres is the sum of the radii

$d = r_1 + r_2 = 5 + 7 = 12$ cm

Answer: Distance between the centres = 12 cm

8. A small sensor lies inside a satellite dish and touches its wall internally. If the dish has radius 15 cm and the sensor has radius 2.5 cm, find the distance between their centres.

For internal contact the distance between the centres is the difference of radii

d = 15 − 2.5 = 12.5 cm

Answer: Distance between the centres = 12.5 cm

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9. An inner holder touches the outer cylindrical container internally. If the outer container has radius 8 cm and the inner holder has radius 6 cm, find the distance between their centres.

$d = R - r = 8 - 6 = 2$ cm

Answer: Distance between the centres = 2 cm

10. A pyramid-shaped sculpture is placed in the centre of a circular plaza with a radius of 10 m. A decorative pole stands 26 m from the centre of the circle. Two guide wires are attached from the pole to the plaza edge, just touching the circle. Find the length of each wire.

$OP = 26$ m, $r = 10$ m, wire PT is a tangent

$PT^2 = OP^2 - r^2 = 26^2 - 10^2$

$PT^2 = 676 - 100 = 576$

$PT = \sqrt{576} = 24$ m

Answer: Length of each wire = 24 m

Remember — tangents and angles

Tangent $\perp$ radius at the point of contact.

Two tangents from an outside point are equal in length.

External contact: $d = r_1 + r_2$. Internal contact: $d = R - r$.

Angle at the centre = 2 × angle at the circumference.

Tangent–chord angle = angle in the alternate segment.